

Designing an End-to-End Business Intelligence Solution

Greenwich Global, Business Intelligence Dashboard

Sales Performance & Customer Demographic Analysis

Tools & Technologies:

Microsoft Power BI (DAX, Power Query, Data Modeling)

Microsoft Excel

ChatGPT (AI-assisted documentation and presentation support)

Developed By:

Comfort Egbebu

[GitHub Profile](#)

[LinkedIn Profile](#)

[Power BI Project](#)

Business Understanding

Greenwich Global Store is a retail business specializing in three major product categories: Furniture, Technology, and Office Supplies.

The company operates across East, West, South, and Central regions of the United States, serving a large and diverse customer base through multiple shipment and delivery modes.

Due to the scale of its operations, the company generates large volumes of transactional, customer, product, and regional sales data on a continuous basis.

The business aims to be the leader in the sectors where it serves the customers and thrives to achieve the best customer satisfaction while attaining sustainable business growth. It intends to leverage the use of data and automation to improve customer experience.



Executive Summary

This project was developed to design and deliver an end-to-end business intelligence solution for Greenwich Global Store using Microsoft Power BI. The objective was to transform raw transactional data into meaningful insights that support strategic business decision-making across sales, profitability, customer behavior, and regional operations.

Using Power BI (DAX, Power Query, and Data Modeling), CSV datasets containing transactional, customer, product, and regional data were cleaned, transformed, modeled, and visualized into an interactive executive dashboard solution.

The solution enables management to monitor key performance indicators (KPIs), identify business growth opportunities, evaluate operational performance, and support data-driven strategic decision-making through dynamic and interactive reporting.

The dashboards provide comprehensive analysis across key business areas including:

- Sales and profit performance
- Profit margin analysis
- Customer demographic insights
- Regional sales performance
- Product profitability trends
- Shipment and delivery analysis
- High-value customer identification

\$470.53K Total Sales

\$61.62K Total Profit

573 Total customers





There is a Problem!

- The absence of a unified analytical reporting system limits the organization's ability to make timely, and data-driven strategic decisions.
- The management lacked an efficient way to monitor key performance indicators (KPIs), evaluate regional sales performance, analyze product profitability, and identify high-value customers across the business.
- Hence, Greenwich Global Store required a centralized reporting solution capable of transforming fragmented raw CSV data into meaningful business insights.
- To address this challenge, an interactive Power BI dashboard solution was developed to support dynamic reporting, performance monitoring, and business analysis.

Project Objectives



The primary objective of this project was to develop an interactive business intelligence solution capable of transforming raw operational data into actionable business insights for strategic decision-making.

Key Objectives:

- Develop an end-to-end business intelligence dashboard using Microsoft Power BI
- Monitor key sales and profitability KPIs
- Analyze regional sales performance across business territories
- Evaluate product-level sales and profit trends
- Identify high-value customers and customer segments
- Analyze shipment and delivery performance
- Enable dynamic and interactive business reporting
- Support data-driven decision-making through visual analytics

Data Overview

This slide focuses on understanding the datasets

Dataset/Table	Purpose
Sales	Transactional sales and profit records
Customer_dim	Customer demographic information
Product_dim	Product categories and details
Order_dim	Order transactional data
Location_dim	Regional sales data
Calendar_dim	Time intelligence and trend analysis

- The project utilized multiple raw CSV datasets containing transactional, customer, product, order, shipment, and regional business data.
- These datasets were imported, cleaned, transformed, and integrated within Microsoft Power BI using Power Query and data modeling techniques to support comprehensive business analysis and reporting.

Order ID	Order Date	Ship Date	Ship Mode	Customer	Customer ID	Segment	Region	Category	Product ID	Product Name	Sales	Quantity	Discount	Profit	Profit Margin
CA-2015-1	1/31/2015	2/5/2015	Second Cla	KB-16315	Karl Braun	Consumer	Central	Office Supp	OFF-PA-10	Xerox 1995	12.96	2	0	6.2208	0.48
CA-2015-1	1/31/2015	2/5/2015	Second Cla	KB-16315	Karl Braun	Consumer	Central	Furniture	FUR-FU-10	Seth Thom	53.34	3	0	16.5354	0.31
CA-2015-1	1/31/2015	2/5/2015	Second Cla	KB-16315	Karl Braun	Consumer	Central	Office Supp	OFF-BI-100	Ibico Stand	32.96	2	0	16.1504	0.49
CA-2015-1	1/2/2015	1/9/2015	Standard C	VW-21775	Victoria Wi	Corporate	East	Office Supp	OFF-LA-10	Avery 505	23.68	2	0.2	8.88	0.375
CA-2015-1	1/2/2015	1/9/2015	Standard C	VW-21775	Victoria Wi	Corporate	East	Furniture	FUR-BO-10	O'Sullivan	452.45	5	0.5	-244.323	-0.54
CA-2015-1	1/2/2015	1/9/2015	Standard C	VW-21775	Victoria Wi	Corporate	East	Technology	TEC-PH-10	Speck Prod	62.982	3	0.4	-14.6958	-0.233333
CA-2015-1	1/2/2015	1/9/2015	Standard C	VW-21775	Victoria Wi	Corporate	East	Technology	TEC-MA-1C	Cisco 9971	1188	9	0.7	-950.4	-0.8
CA-2015-1	1/2/2015	1/9/2015	Standard C	VW-21775	Victoria Wi	Corporate	East	Technology	TEC-AC-10	Sony Micro	89.584	2	0.2	4.4792	0.05
US-2015-1	1/17/2015	1/24/2015	Standard C	JS-15940	Joni Sundai	Home Office	Central	Furniture	FUR-FU-10	Eldon Clust	254.744	7	0.6	-312.0614	-1.225
US-2015-1	1/10/2015	1/15/2015	Standard C	MM-17920	Michael Mc	Consumer	East	Furniture	FUR-TA-10	Bush Advan	1018.104	4	0.4	-373.3048	-0.366667
CA-2015-1	1/30/2015	2/4/2015	Standard C	JP-15520	Jeremy Pisi	Consumer	Central	Office Supp	OFF-PA-10	Rediform V	14.304	6	0.2	5.0064	0.35
CA-2015-1	1/23/2015	1/27/2015	Standard C	WW-11905	Carl Weiss	Home Office	West	Office Supp	OFF-PA-10	Wirebound	29.04	3	0	13.9392	0.48
CA-2015-1	1/23/2015	1/27/2015	Standard C	WW-11905	Carl Weiss	Home Office	West	Office Supp	OFF-LA-10	Avery 500	14.62	2	0	6.8714	0.47
CA-2015-1	1/2/2015	1/4/2015	Second Cla	CB-12535	Claudia Ber	Corporate	East	Office Supp	OFF-ST-10	SAFCO Mol	85.52	2	0	22.2352	0.26
CA-2015-1	1/2/2015	1/4/2015	Second Cla	CB-12535	Claudia Ber	Corporate	East	Office Supp	OFF-AR-10	Newell 313	9.84	3	0	2.6568	0.27
CA-2015-1	1/2/2015	1/4/2015	Second Cla	CB-12535	Claudia Ber	Corporate	East	Office Supp	OFF-PA-10	Xerox 1955	20.04	3	0	9.6192	0.48
CA-2015-1	1/27/2015	1/29/2015	Second Cla	SV-20365	Seth Verno	Consumer	East	Furniture	FUR-CH-10	High-Back I	181.986	2	0.3	-54.5958	-0.3
CA-2015-1	1/27/2015	1/29/2015	Second Cla	SV-20365	Seth Verno	Consumer	East	Technology	TEC-PH-10	Polycom V	431.976	4	0.4	-100.7944	-0.233333
CA-2015-1	1/27/2015	1/29/2015	Second Cla	SV-20365	Seth Verno	Consumer	East	Technology	TEC-PH-10	Nortel Mer	155.37	1	0.4	-36.253	-0.233333
CA-2015-1	1/30/2015	2/6/2015	Standard C	CC-12145	Charles Cre	Consumer	West	Furniture	FUR-FU-10	Computer	227.36	7	0	81.8496	0.36
CA-2015-1	1/30/2015	2/6/2015	Standard C	CC-12145	Charles Cre	Consumer	West	Technology	TEC-MA-1C	Bady BDG1	1919.976	3	0.2	215.9973	0.1125
CA-2015-1	1/3/2015	1/8/2015	Standard C	CSF-20965	Sylvia Foul	Corporate	Central	Office Supp	OFF-PA-10	Xerox 21	10.368	2	0.2	3.6288	0.35
CA-2015-1	1/5/2015	1/10/2015	Standard C	CSR-20740	Steven Roe	Home Office	West	Furniture	FUR-CH-10	Global Del	61.584	1	0.2	-6.9282	-0.1125
CA-2015-1	1/13/2015	1/17/2015	Standard C	CPB-19150	Philip Brow	Consumer	West	Office Supp	OFF-BI-100	Binding Ma	70.008	3	0.2	24.5028	0.35
CA-2015-1	1/13/2015	1/17/2015	Standard C	CPB-19150	Philip Brow	Consumer	West	Furniture	FUR-FU-10	Floodlight I	77.6	4	0	38.024	0.49
CA-2015-1	1/13/2015	1/17/2015	Standard C	CPB-19150	Philip Brow	Consumer	West	Furniture	FUR-FU-10	Deflect-o E	464.85	9	0	92.97	0.2
CA-2015-1	1/3/2015	1/8/2015	Second Cla	AH-10465	Amy Hunt	Consumer	Central	Technology	TEC-AC-10	Emermax A	398.4	5	0.2	84.66	0.2125
CA-2015-1	1/3/2015	1/8/2015	Second Cla	AH-10465	Amy Hunt	Consumer	Central	Office Supp	OFF-AR-10	Newell 317	7.056	3	0.2	0.7938	0.1125
CA-2015-1	1/3/2015	1/8/2015	Second Cla	AH-10465	Amy Hunt	Consumer	Central	Furniture	FUR-BO-10	O'Sullivan I	1352.398	9	0.32	-437.5404	-0.323529
CA-2015-1	1/12/2015	1/18/2015	Standard C	YS-21880	Yana Soren	Corporate	East	Office Supp	OFF-ST-10	Dual Level	465.18	3	0	120.9468	0.26
CA-2015-1	1/17/2015	1/22/2015	Standard C	CSC-20260	Scott Cohe	Corporate	East	Office Supp	OFF-AR-10	Avery Fluoi	6.68	2	0	2.004	0.3

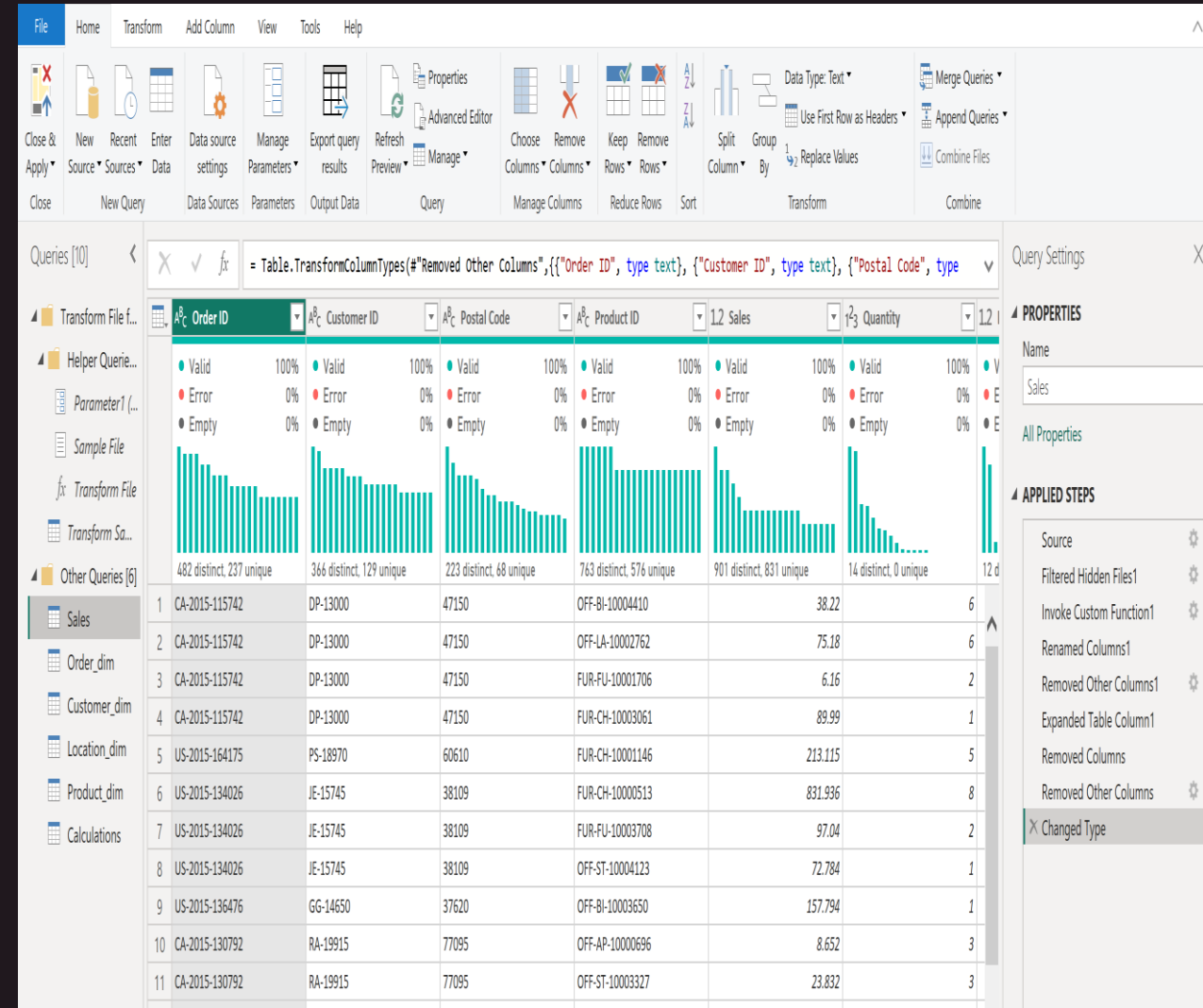
Raw transactional dataset containing sales, customer, shipment, geographic, and profitability information

Data Cleaning & Transformation

Raw business datasets were cleaned, transformed, and prepared in Microsoft Power BI using Power Query to ensure data consistency, accuracy, and analytical readiness.

Data Preprocessing Steps:

- Removed duplicate and irrelevant records
- Corrected data types and formatting inconsistencies
- Handled missing and incomplete values
- Standardized categorical fields and naming conventions
- Prepared datasets for relational modeling and dashboard integration



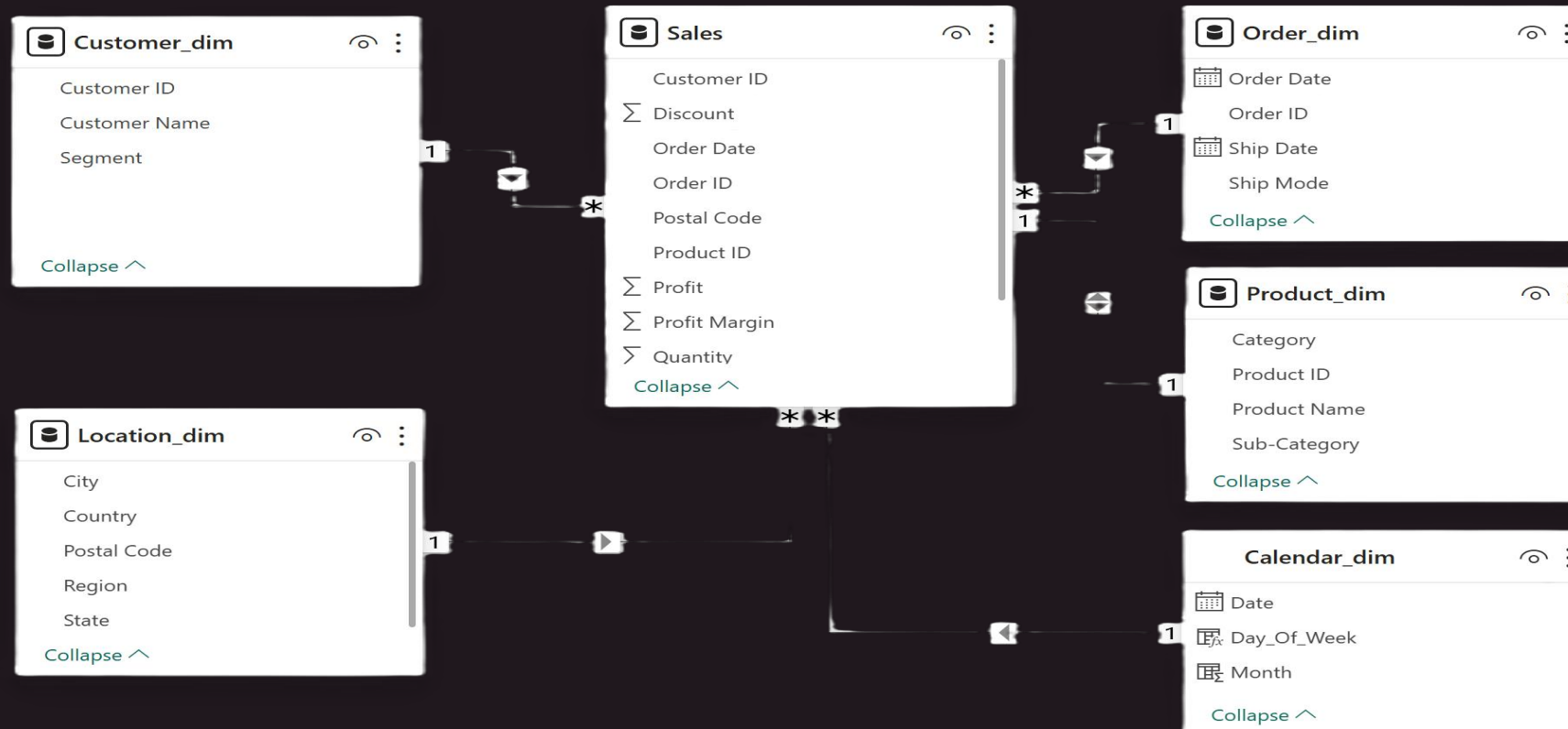
Data Modeling

Power BI relational modeling

The model was designed using a dimensional/star-schema structure to improve analytical efficiency and reporting performance.

A relational data model was developed within Microsoft Power BI by establishing relationships between fact and dimension tables.

This modeling approach enabled dynamic filtering, accurate KPI calculations, time intelligence analysis, and interactive dashboard reporting across multiple business dimensions.



DAX Measures & KPI Calculations

DAX measures and calculated KPIs were developed to support dynamic business analysis, profitability evaluation, and interactive dashboard reporting.

KPI / Measure	Purpose
Total Sales	Measures overall revenue performance
Total Profit	Evaluates business profitability
Profit Margin %	Assesses profitability efficiency
Sales Growth Trend	Measures sales performance overtime
Average Order Value	Measures customer spending behavior
Total Orders	Tracks transaction volume
Customer Count	Shows total number of customers the business has

Dashboard Overview

Two interactive dashboards were developed in Microsoft Power BI to support sales performance monitoring and customer demographic analysis. The dashboards provide dynamic KPI tracking, regional analysis, profitability insights, and customer-focused business intelligence reporting. See next slides.

Greenwich Sales Performance Dashboard

Sales

Demography

\$470.53K

Total Sales

\$61.62K

Total Profit

13.10%

Profit Margin %

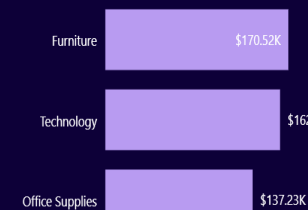
2,102

Total Orders

\$453.3

Avg Order Value(AOV)

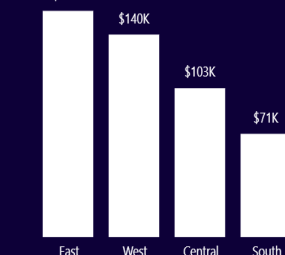
Sales by Category



Top 10 Most Profitable Products By Region



Sales by Region



Sales Trend



Greenwich Customer Demographic Analysis Dashboard

Sales

Demography

573

Total Customers

7,979

Total Quantity Sold

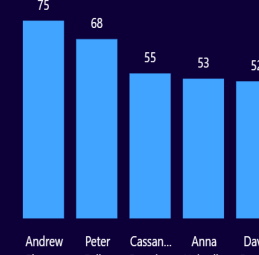
1,038

Total Unique Orders

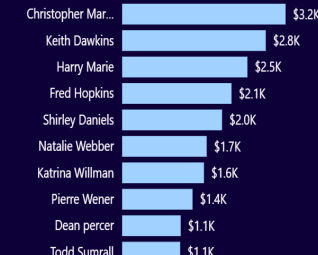
\$821.2

Average Sales per Customer

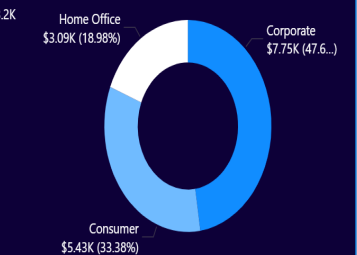
Top 5 Active Customers



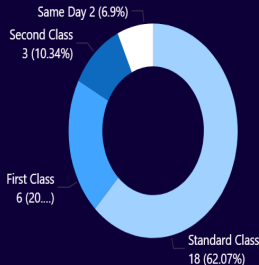
Top 10 Profitable Customers



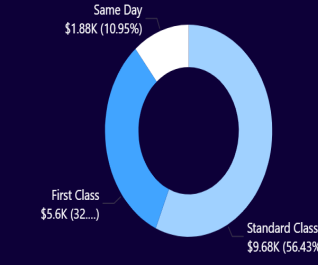
Profit by Customer Segment



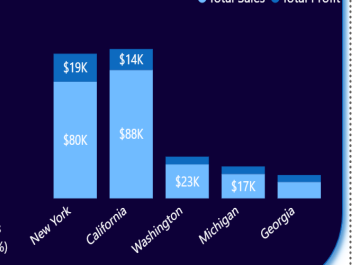
Customers Preferred Shipment Mode



Profit by Shipment Mode



Profit By States



Greenwich Sales Performance Dashboard

Sales

Demography

\$470.53K

Total Sales

\$61.62K

Total Profit

13.10%

Profit Margin %

2,102

Total Orders

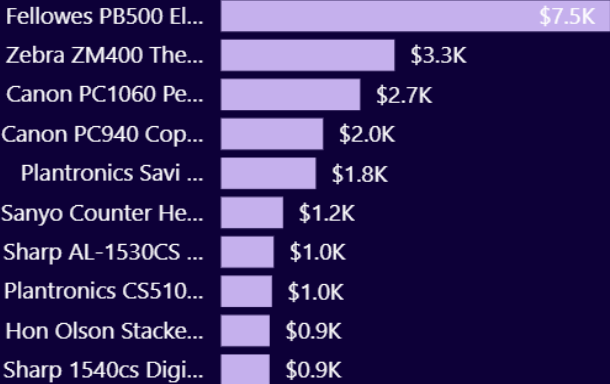
\$453.3

Avg Order Value(AOV)

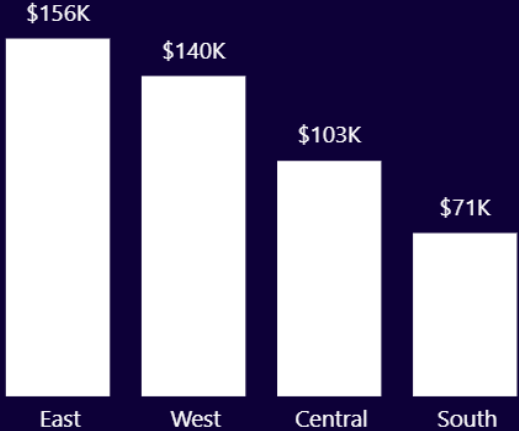
Sales by Category



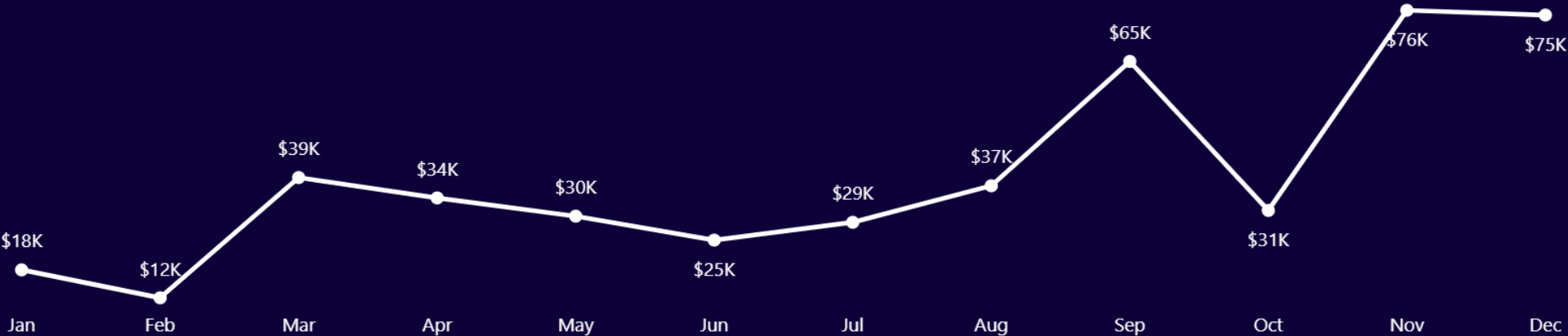
Top 10 Most Profitable Products By Region



Sales by Region



Sales Trend



Greenwich Customer Demographic Analysis Dashboard

Sales

Demography

573

Total Customers

7,979

Total Quantity Sold

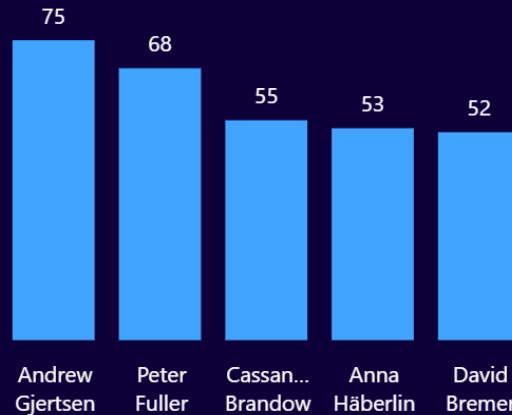
1,038

Total Unique Orders

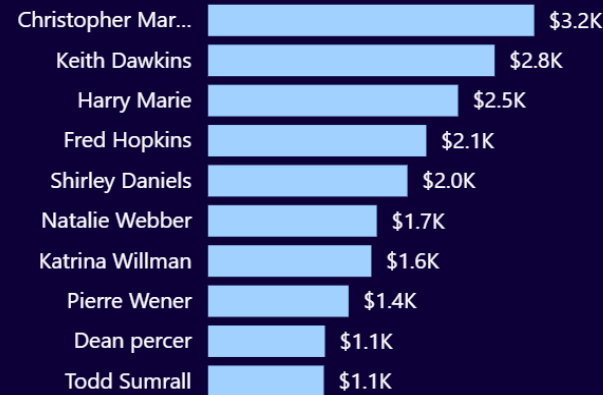
\$821.2

Average Sales per Customer

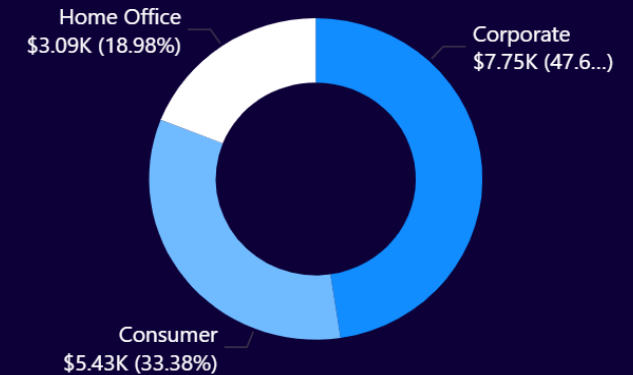
Top 5 Active Customers



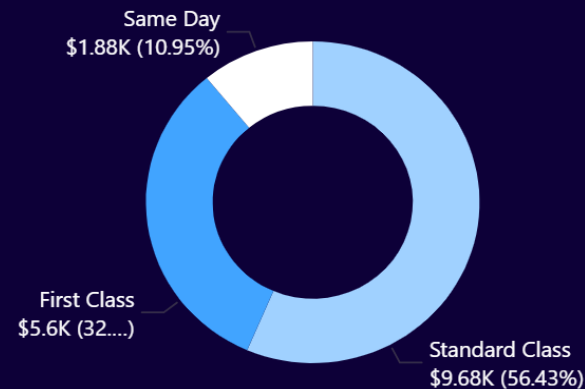
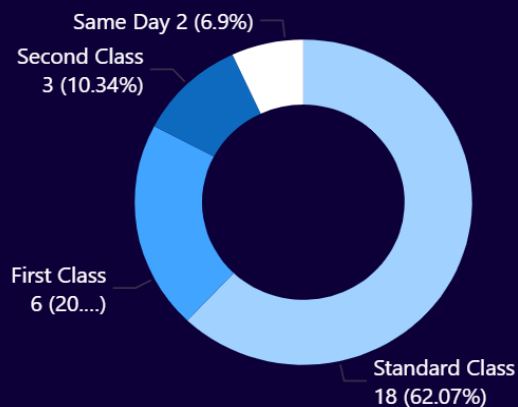
Top 10 Profitable Customers



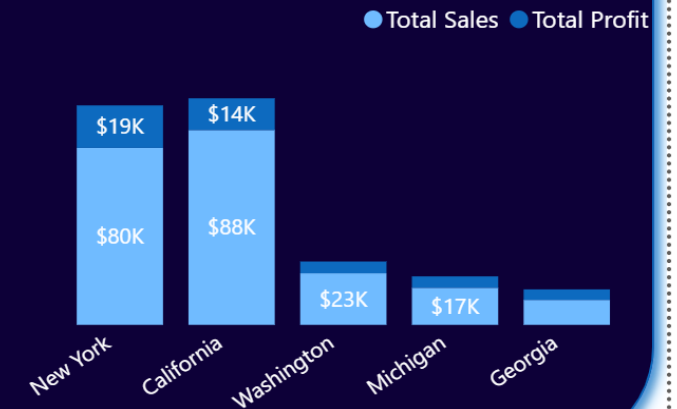
Profit by Customer Segment



Customers Preferred Shipment Mode Profit by Shipment Mode



Profit By States



Sales Dashboard Analysis

The Sales dashboard focuses on business performance

Key Business Insights / Findings

Greenwich Sales Performance Dashboard

Sales

Demography

\$470.53K

Total Sales

\$61.62K

Total Profit

13.10%

Profit Margin %

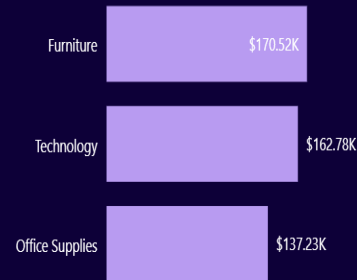
2,102

Total Orders

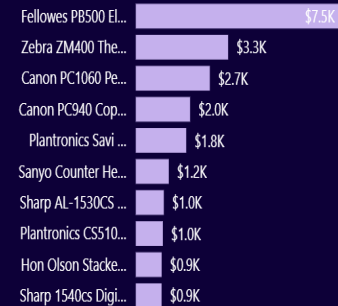
\$453.3

Avg Order Value(AOV)

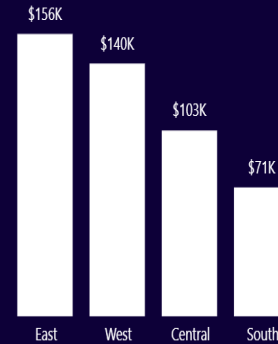
Sales by Category



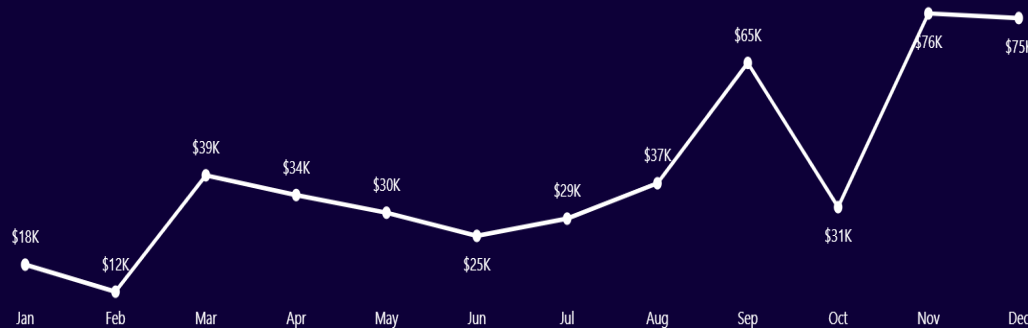
Top 10 Most Profitable Products By Region



Sales by Region



Sales Trend



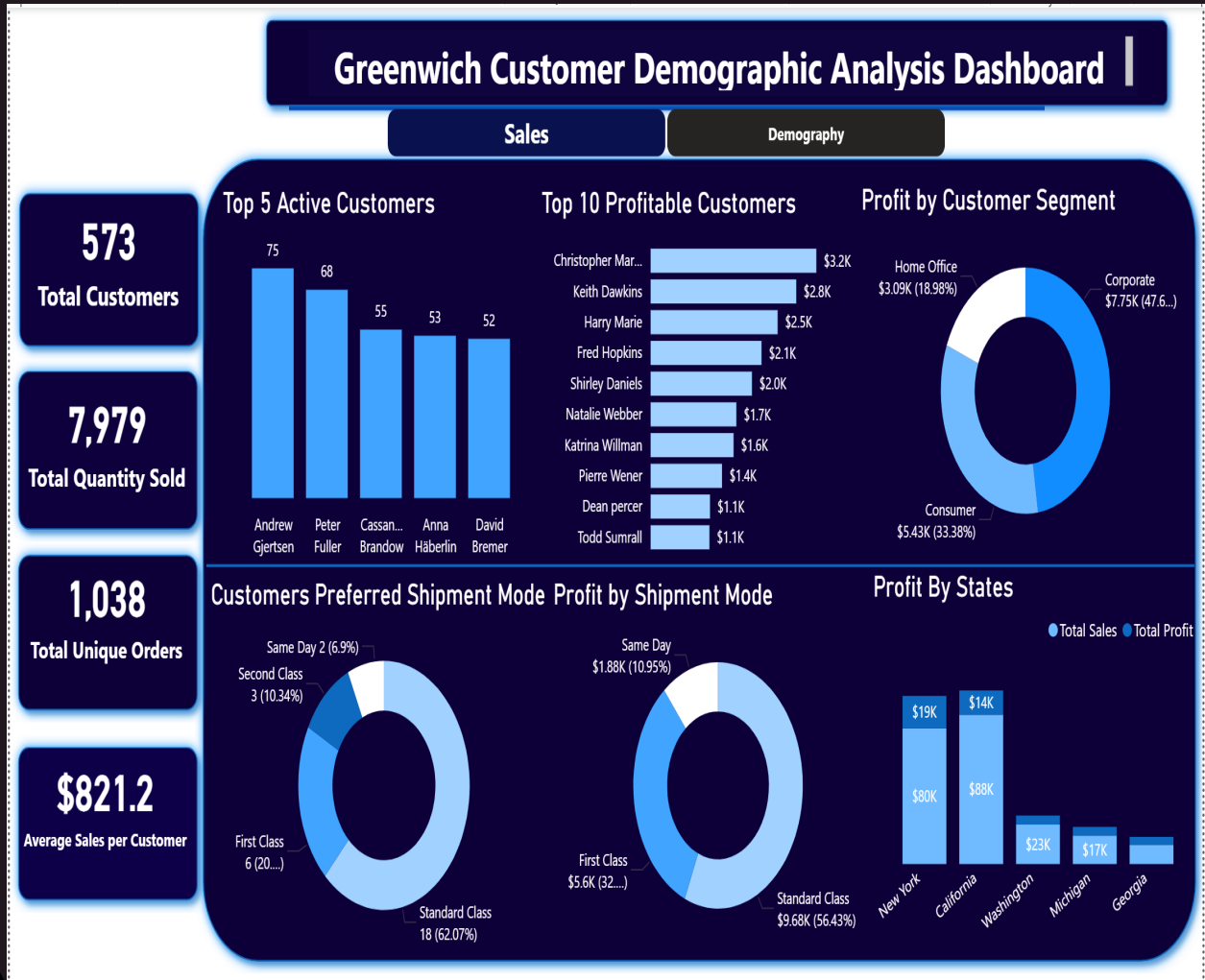
- Furniture category generated the highest overall sales revenue contribution of \$170.52K. Office Supplies recorded the least sales of \$137.23K
- East region generated the highest overall sales performance, contributing approximately \$156K in revenue, and South region recorded the lowest overall sales contribution of \$71K
- Sales performance varied significantly across regions and product categories.
- Certain products generated high sales volumes but relatively lower profit margins.
- Standard Class shipment mode accounted for the majority of product deliveries.
- Profitability trends revealed uneven margin performance across product segments.

Actionable Insights / Recommendations (Sales):

- Despite generating highest sales revenue from Furniture, deeper analysis revealed that much more profit was made from the sales of Technology. This clearly shows that high sales do not equal high profit.
- The strength of the business lies within the Technology sector where it has higher profit margins. Therefore, the business should focus more on Technology while working hard to improve other sectors.
- Management should evaluate the possible factors contributing to low profits in Furniture. These may include; high operational costs, expensive logistics and weaker pricing, among others.
- Management should review the pricing strategies, discount policies and cost structures within the Furniture category to improve profit margins. Product profitability analysis should be prioritized alongside sales performance to support more effective business decision-making.

Customer Demographic Analysis

Analysing customer's behavioural patterns

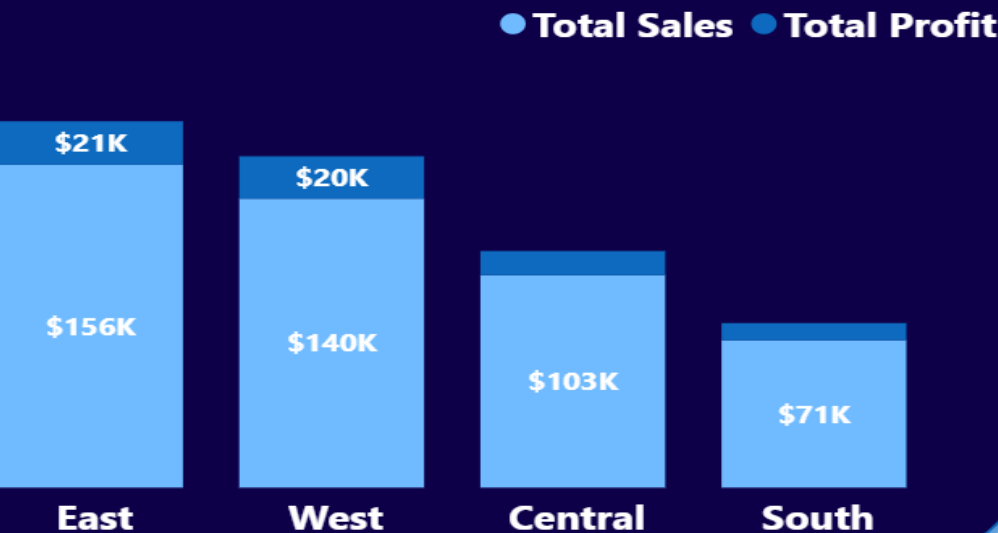


Key Customer Insights

- The business served a total of 573 customers across multiple customer segments and regions.
- Corporate customers generated the highest overall profitability contribution among all customer segments.
- Standard Class shipment mode accounted for the majority of customer deliveries and profit generation.
- A small group of customers generated higher profitability despite lower transaction volumes, indicating the presence of high-value customers.
- New York and California recorded the strongest combined sales and profitability performance among all states.
- Customer purchasing behavior and profitability varied significantly across shipment modes and customer segments.
- Regional customer distribution revealed concentration in the East and West regions

Regional Performance Analysis

Sales vs Profit by Region



Active Customers by Region



Comparative Business Insights

Region	Active Customers	Sales	Profit	Profit Margin
West	276	\$140K	\$20K	14.64%
East	246	\$156K	\$21K	13.49%
Central	201	\$103K	\$11.7K	11.39%
South	155	\$71K	\$8.3K	11.66%

Key Regional Insights



- Despite having fewer customers, lower profit margin (when compared to West), East generated the highest sales revenue (\$156K) and highest profit(\$21K). The interpretation is that East customers spend more on average and generate stronger overall business value.
- West achieved the highest profit margin (14.64%), highest active customer count, second highest sales and second highest profit. The business benefit is that West operates most efficiently, converting sales into profit better than any other region.
- Central region had moderate customer base, moderate sales and profit. This signals a stable region with potentials for growth.
- The Southern region experienced the lowest customer count, lowest sales and profit. But surprisingly has higher profit margins than the Central. Indicating an underperforming market with opportunities for growth.

Overall, the region with the most customers was not the region with the highest sales or profit, suggesting that customer quality and purchasing behavior are more important than customer quantity alone.

Actionable Insights / Recommendations

- East region generated the highest sales revenue and total profit but did not achieve the highest profit margin. Further analysis of product mix, discounting patterns, pricing strategies, and operational costs is recommended to identify opportunities for improving profitability efficiency. The business may consider potential initiatives such as pricing optimisation, product bundling, cross-selling, and operational efficiency measures to enhance margins.

These strategies will complement the efforts for customer acquisition and increase customer value and spending behaviours.

- Best practices contributing to West region's strong profit margin should be evaluated and replicated across other regions.
- Management should investigate customer purchasing behaviour, average transaction value, and product mix within the East region to identify best practices that can be replicated across other territories.
- South and Central regions present opportunities for targeted growth initiatives to improve customer engagement and profitability.

Executive Recap : Key Findings Summary

Sales Performance

- Total sales reached \$470.53K across all regions.
- Total Profit made was \$61.62K
- The business operated at 13.10% Profit margin
- The business offered three Product categories; furniture, technology and office supplies.
- Furniture generated the highest sales revenue contribution (\$170K).
- Technology generated the highest profit contribution (\$33.5K).
- Trend analysis showed a rise in sales across all regions in the last quarter of the year starting September with the peak being November. A sharp decline was observed in Oct.
- Sales performance increased significantly during the final quarter, with November recording the highest sales activity.

Regional Performance

There were four Regions of operation; East, West, Central and South.

- East region generated the highest sales (\$156K) and total profit (\$21K).
- West region achieved the highest profit margin (14.64%), indicating superior profitability efficiency

Customer Insights

- The business served a Total number of 573 customers.
- West region recorded the highest active customer count (276).
- Higher customer volume did not necessarily result in higher profitability.

Operational Insights

- Standard Class was the most frequently used shipment mode and generated the highest shipment-related profit contribution for the business.

Skills & Competencies Demonstrated

Business Intelligence Skills

- Business Requirements Analysis
- KPI Development
- Dashboard Design
- Data Storytelling
- Performance Analysis

Power BI Skills

- Power Query Transformations
- Data Modeling
- Star Schema Design
- DAX Measures & Calculations
- Interactive Dashboard Development



Analytical Skills

- Sales Analysis
- Profitability Analysis
- Customer Segmentation
- Regional Performance Evaluation
- Business Recommendation Development

Tools Used

- Microsoft Power BI
- DAX
- Power Query
- Microsoft Excel
- Artificial Intelligence



Conclusion

- Greenwich Global Store should focus on expanding its customer base while simultaneously increasing customer value through targeted retention and profitability initiatives. This approach can support sustainable revenue growth, margin improvement, and long-term business performance.
- Conclusively, this project successfully transformed raw retail transaction data into an interactive business intelligence solution capable of supporting strategic decision-making.
- Through data cleaning, modeling, KPI development, and dashboard design, key insights were uncovered regarding sales performance, profitability, customer behavior, and regional efficiency.
- The solution enables stakeholders to monitor business performance dynamically and identify opportunities for growth, profitability improvement, and operational optimization..



Thanks!



Let's talk business...